

ReForge AI

MSME IDEA HACKATHON 6.0 Phase 2 Application

1. Team Members Name & Aadhar Card No.

- a) Aditya Bhowmik
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1.14.1 Name and details of Mentors *(Maximum 500 characters)*

Prof. Dr. Sudipta Sahana, Head of the Department, Computer Science & Engineering (AI), Institute of Engineering & Management (IEM) – University of Engineering & Management (UEM), Kolkata. He is the mentor guiding our team for the MSME Idea Hackathon 2026 project.

1.14.2 Experience and Qualification of Mentors

Qualification: Post Doctorate – Asian Institute of Technology (Thailand); Ph.D. – University of Kalyani; B.Tech. & M.Tech. – WBUT.

Experience: Over 16 years of teaching and research experience. Published 100+ research papers in international journals and conferences. Research interests include Cloud Computing and Machine Learning. Life Member of CSI and Fellow Member of IETE.

1.14.3 Contact Details of Mentors

Name: Prof. Dr. Sudipta Sahana

Designation: Head of the Department, CSE (AI), IEM-UEM Kolkata

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2.2 Whether the idea involves use of existing intellectual property or not, give brief detail thereof *(Maximum 1500 characters)*

The proposed solution does not intentionally use or reproduce any third-party proprietary intellectual property. It combines existing publicly available technologies and software frameworks, including computer vision, multimodal AI, parametric 3D geometry generation and web-based 3D rendering, into an integrated reverse-engineering workflow. The proposed innovation lies in the system-level integration and workflow for converting multi-view images of industrial components into structured engineering parameters, an interactive parametric 3D representation, and an AI-assisted engineering analysis interface. Any third-party libraries, APIs or models used during implementation will be used in accordance with their respective licenses and terms.

2.7 Summary of the idea *(Maximum 750 characters)*

ReForge AI is an AI-assisted reverse-engineering system for industrial components and hard-to-source spare parts. It analyzes multiple images of a physical component using multimodal AI to identify its type, features and engineering parameters, then converts the extracted parameters into an interactive parametric 3D reconstruction. The system presents dimensions, material and manufacturing insights alongside the model and provides an AI Engineer interface for technical queries. The key advancement is the integration of image-based component analysis, parameter extraction, dimension-driven 3D reconstruction and engineering assistance into a single workflow, reducing dependence on manual reconstruction and CAD modelling.

2.8(a) Is it a new concept?

YES

2.8(b) Prior art on the concept, if any

Existing prior art includes conventional reverse-engineering workflows involving physical measurement, CAD reconstruction, 3D scanning and computer-aided inspection. Image-based 3D reconstruction and AI-based computer vision are also established areas. However, ReForge AI focuses on integrating multi-view image analysis, engineering parameter extraction, dimension-driven parametric reconstruction and an AI-assisted engineering interface into a unified workflow for industrial spare/component reverse engineering.

2.9 Main Problem Being Addressed in the Project *(Maximum 500 characters)*

Industrial machinery often depends on obsolete, imported or unavailable spare parts. Replacing such components can involve high procurement costs, long lead times and dependence on external suppliers. When drawings or CAD files are unavailable, reconstructing a replacement component requires manual measurement, engineering judgement and CAD modelling. ReForge AI aims to reduce this effort by converting images of an existing component into structured engineering information and an interactive parametric 3D reconstruction.

2.10(a) Who is it for? *(Maximum 300 characters)*

The system is intended for manufacturing industries, maintenance teams, mechanical engineers, machine shops, MSMEs and organizations that need to reproduce or replace obsolete, imported or unavailable mechanical components without readily available CAD drawings.

2.10(b) What will it do? *(Maximum 300 characters)*

ReForge AI analyzes multiple images of a physical component, identifies its geometry and features, estimates engineering parameters, generates an interactive parametric 3D reconstruction, and provides material, manufacturing and engineering insights through an AI-assisted interface.

2.10(c) Any unique features? Explain. *(Maximum 300 characters)*

Its key feature is a unified image-to-engineering workflow: multi-view AI analysis produces structured parameters that directly drive a dimension-aware 3D reconstruction. The same engineering data is reflected in the model, measurements and AI Engineer interface, creating a traceable reconstruction workflow.

2.11 How simple or complex will the idea's execution or implementation be? What are the risk factors involved in executing the idea? *(Maximum 500 characters)*

The prototype involves multimodal AI, image processing, parametric 3D geometry generation and web-based 3D visualization, making implementation moderately complex. Major risks include inaccurate dimensional estimation from images, insufficient visual information, component variations and incorrect AI interpretation. Reference dimensions, confidence scores, uncertainty reporting and parameter validation will be used to reduce these risks.

2.12 How soon could the idea be put into operation? (TRL of prototype) *(Maximum 200 characters)*

A functional software prototype can be developed within approximately 2–3 months. The initial prototype will demonstrate image analysis, parametric 3D reconstruction and AI-assisted engineering interaction.

2.13 How much investment would you need for prototyping of the Idea? *(Maximum 200 characters)*

An estimated investment of ₹2 lakh would be required for initial prototyping, covering AI/API usage, software development, 3D reconstruction implementation, testing with sample industrial components, and basic cloud and deployment requirements.

2.14(a) How do you intend to protect your idea (i.e. your intellectual property or IP)? Status of IPR (If any) *(Maximum 300 characters)*

The project is currently at the prototype-development stage and no patent/IPR has been filed yet. Our team intends to evaluate patentability of the integrated reverse-engineering workflow and protect eligible technical innovations through appropriate IPR mechanisms before public disclosure.

2.14(b) Related Background *(Maximum 500 characters)*

Existing approaches include manual reverse engineering using physical measurements and CAD software, 3D scanning and metrology-based reconstruction, photogrammetry and computer-vision-based 3D reconstruction, and AI-based image understanding. These approaches generally address individual stages of the workflow. ReForge AI focuses on integrating visual component analysis, engineering parameter extraction, parametric reconstruction and AI-assisted engineering interpretation into one workflow.

2.15 Current Development Status of innovation *(Maximum 500 characters)*

The project is currently in the prototype development stage. The frontend workflow has been designed, including the landing page, multi-image upload interface and reconstruction interface. Development is underway for the Gemini-based component analysis, dimension-driven parametric 3D reconstruction and AI Engineer interface. The initial prototype is being developed to validate the complete image-to-engineering workflow.

2.16 Expected time of completion of idea development *(Maximum 500 characters)*

The functional prototype is expected to be completed within approximately 2–3 months. The initial development will focus on multi-image component analysis, structured engineering parameter extraction, interactive parametric 3D reconstruction and AI-assisted engineering interaction. Further development will expand supported component types, improve dimensional reliability and validate the system on real industrial components.

2.19 How is this project made and used? *(Maximum 500 characters)*

An MSME or product engineer uploads multiple images of a damaged or unavailable part and enters known dimensions. ReForge AI identifies its geometry, features, material and manufacturing requirements, then generates an interactive 3D reconstruction. The engineer reviews dimensions and recommendations, queries the AI Engineer, and selects a suitable process. The platform then matches required materials and manufacturing capabilities with local MSMEs, foundries, tool rooms and machine shops for sourcing and production.

3.1 Total idea/project cost (Rs in lakh)

No.	Cost Head	Estimated Cost	Details
1	AI/software development and engineering	₹3.00 lakh	Application development, multimodal AI integration, backend/API development, testing and engineering effort.
2	3D reverse-engineering and parametric reconstruction	₹1.50 lakh	Parametric geometry engine, interactive 3D visualization, dimensional mapping and reconstruction workflow.
3	Cloud, API and compute infrastructure	₹0.75 lakh	AI API usage, cloud hosting, inference/compute and development infrastructure during prototyping and testing.
4	Industrial scanning/metrology access	₹1.25 lakh	Access to 3D scanning, dimensional measurement and metrology facilities for calibration and validation of reconstructed components.
5	Prototype hardware and component testing	₹1.00 lakh	Sample industrial components, test fixtures, measurement accessories and prototype validation activities.
6	Industrial datasets and domain	₹0.75 lakh	Collection/curation of component images, engineering

No.	Cost Head	Estimated Cost	Details
	validation		references and validation across representative mechanical components.
7	Security, deployment and industrial validation	₹0.75 lakh	Deployment hardening, access controls, monitoring, system validation and pilot-readiness activities.
8	Contingency and unforeseen development costs	₹1.00 lakh	Allowance for additional testing, API usage, hardware/access requirements and unexpected development needs.
	TOTAL PROJECT COST	₹10.00 lakh	Indicative total development and pilot-validation budget

3.1 Upload Block diagram/flow chart/Circuit Diagram/Pictures

Separate PDF required has been attached for the block diagrams

4. Is this Idea recommended by HI?

Yes

5. Whether the verification of the age of Incubatee carried out by HI.

Yes